



Task 8 - Individual Reflective Report

Group Project Spring 2026

Paul Brocvielle

Montrésor Group

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1. Context and Task 1 Baseline

This report is my individual Task 8 reflection for the group project. It looks back at the role I proposed in Task 1, how I actually contributed across the project, and what the experience showed about my working style in a technical engineering team.

In Task 1 I described myself as an ENTP-A “Debater”. The main traits I identified were curiosity, creativity, adaptability, confidence under pressure, and a preference for understanding how systems work before accepting a fixed solution. My strongest team roles were **Innovator, Evaluator, and Teamworker**, with **Investigator** as a secondary strength. Based on that profile, I proposed the role of **Technical Strategist & System Designer**. In Task 2, the group formalised this as **Technical Strategist & System Architect**.

The practical promises attached to that role were clear: propose and compare technical approaches, design the overall architecture, support debugging and optimisation, challenge weak design choices constructively, and contribute external research. The project provides a useful test of those claims, moving from algorithm comparison in Tasks 3-4 to applied cooling-tower optimisation in Task 6 and structural validation in Task 7.

My high-level conclusion is that Task 1 was directionally accurate. I brought value through technical reasoning, questioning assumptions, workflow setup, and reproducibility. However, idea generation and architecture thinking are not enough on their own. In late phases, where finalisation and visible ownership mattered most, my contribution was less visible than in earlier stages.

2. Hindsight on Personality Fit

Several parts of the ENTP-A profile were accurate. I naturally engaged with the “why” behind technical decisions. In Tasks 3 and 4, this matched the need to compare Simulated Annealing, Particle Swarm Optimisation, and BFGS not only as code outputs, but as different search behaviours. I was comfortable questioning whether a result was meaningful, whether an algorithm comparison was fair, and whether the workflow was reproducible enough for someone else to rerun.

The Innovator and Evaluator parts of my profile also appeared in project structure work. Around Tasks 3-4, I helped turn the project from separate scripts and generated outputs into a cleaner workflow with setup scripts, dependency files, a clearer README, and ignored generated artifacts. That is not the most visible academic content, but it matters because reproducibility is part of technical quality.

Where Task 1 was incomplete was on follow-through and final-stage visibility. I had identified that I move quickly and find detailed finishing less natural than conceptual work. By Tasks 6-7, the work required sustained reporting, validation, and final integration. Achille and Eneko had more visible ownership in those loops, while my contribution was more supportive. The original profile was not wrong; I simply need to convert strategic input into clearer, owned deliverables more consistently.

3. My Contribution Across the Project

3.1 Early role setup and technical direction

In the first project phase, my contribution was mainly conceptual. Task 2 records that Paul and Aiert were expected to lead methodological research for Tasks 3-4, while Achille coordinated the group and Eneko supported quality analysis. This matched my proposed role: I was useful in discussions about how to compare methods fairly, structure benchmarks, and explain algorithm behaviour. The technical direction needed more than running algorithms — it required connecting code behaviour, mathematical assumptions, and engineering judgement.

3.2 Tasks 3 and 4: optimisation strategy and experiment interpretation

Task 3 compared SA and PSO on benchmark problems, emphasising global search capability, precision, constraint handling, repeated runs, and equal evaluation budgets. My role aligned with technical strategy: discussing what a fair comparison should mean and keeping the work focused on interpretable evidence.

Task 4 added BFGS and made the comparison more interesting. The key interpretation: BFGS was extremely efficient on Rosenbrock but fragile on Rastrigin because local gradient information can trap it in local minima. The lesson: no single method was universally best. The final recommendation of a hybrid strategy — global searcher before BFGS local refinement — matched the architectural reasoning I expected to provide in Task 1.

3.3 Reproducibility, workflow, and repository structure

My clearest repo-visible contribution is the early February 2026 work on project setup. Commit 75bef48 added cross-platform setup scripts, a `requirements.txt`, a `.gitignore`, and a clearer README with project structure and run instructions. This supported the whole team because the project had multiple scripts, generated results, and environment assumptions.

I also contributed to output-path and plotting robustness (073e601), updating plotting helpers so output directories could be passed explicitly instead of being hard-coded. This affected 3D plots, trajectory plots, and Task 3/4 experiment scripts. I also cleaned the repository by removing generated artifacts (8ab3331) and adding `__pycache__` handling (cb0ec5f). This is modest compared with building a full technical model, but aligns with my stated DevOps and system-structuring strengths.

3.4 Task 6 and Task 7 support

Task 6 was the cooling-tower optimisation stage. The group moved from mathematical benchmarks to a real engineering formulation: frustum area and volume, volume constraints, hyperboloid shape constraints, engineering feasibility, and comparison across eight scenarios. Eneko had strong ownership in geometry, plotting, and quality checks; Achille contributed heavily to report integration. My contribution was lighter than in Tasks 3-4 but still architectural: discussing scenario logic, method comparison, and how BFGS/SA/PSO results should be interpreted under constraints.

Task 7 added Abaqus wind-loading analysis, requiring detailed simulation workflow, structural ranking, and careful distinction between engineering-feasible, mathematical fallback, and warning cases. My direct repo-visible ownership was limited; my support was mainly through technical review, ensuring the link from Task 6 optimisation to Task 7 structural interpretation stayed coherent. If

repeating the project, I would claim a defined subtask in the Task 7 pipeline rather than staying in advisory mode.

4. Group Performance and Agile Review

The group used Agile in a practical rather than textbook way. Task 2 established Notion, Kanban-style tracking, a Gantt view, weekly stand-ups, and role allocation. This helped the team start with clarity and gave us a shared language for ownership, deadlines, and progress.

What worked best was iteration. By Tasks 6-7, we had a repeated cycle of generating results, checking plots, correcting interpretation, and rebuilding reports. This mattered because outputs were not simple: a cooling-tower shape could be mathematically compliant but not engineering-feasible; an Abaqus ranking could identify a winner but still have limited convergence confidence. Iteration helped us make those distinctions clearly.

The main weakness was ownership traceability. Roles became fluid — necessary as deadlines and complexity increased — but the group did not always explicitly reassign responsibilities. Some people took strong ownership of final integration while others, including me, contributed in less visible ways. Documentation also lagged code and results at times. For a future Agile project, I would keep the flexible cross-support model but add a lightweight weekly ownership log: who owns which artifact, what “done” means, and what evidence proves completion.

5. Role Evolution: Planned vs Real

My planned role was Technical Strategist & System Designer. My real role was closer to technical strategist plus reproducibility and workflow support, with less late-stage integration ownership than expected.

This evolution had benefits: I contributed where my strengths were strongest — conceptual comparison, method interpretation, architecture, and repository usability. The setup scripts, dependency file, README work, and output-path cleanup all made the project easier to run and maintain.

The cost was that my role became less visible during Task 6-7 finalisation. The late phases rewarded sustained artifact ownership: reports, figures, Abaqus outputs, LaTeX pipelines, final PDFs. I supported the technical direction but did not always translate that into a clearly owned deliverable. This is the main difference between Task 1 expectation and Task 8 reality.

6. Lessons Learned and Career Transfer

First lesson: technical strategy must end in a deliverable. Good ideas, comparisons, and architecture discussions are valuable, but in a team assessment they need to be attached to visible artifacts. When I propose a direction, I will also claim a concrete output or decision record.

Second lesson: reproducibility is engineering work. Setup scripts, dependency management, clean repository structure, and deterministic outputs make technical results more credible and easier to validate. This transfers directly to optimisation, design engineering, and CAE workflows.

Third lesson: role fluidity needs explicit tracking. Flexibility can reduce accountability if ownership is not updated. In future projects I would keep adaptability but pair it with short written handovers and weekly ownership checkpoints.

Fourth lesson (personal): I need to improve finalisation discipline. My natural strength is early-stage technical thinking. My development target is to stay equally engaged when work becomes repetitive, detailed, and close to submission — in engineering, final traceability matters as much as the initial idea.

7. Peer Assessment (0-5)

7.1 Scoring rubric

Each member is scored using the common Task 8 criteria:

- Technical Quality (25%)
- Delivery Reliability (20%)
- Integration and Handover Quality (15%)
- Problem-Solving Under Constraints (15%)
- Collaboration Quality (15%)
- Adaptability and Learning (10%)

7.2 Scores table

Member	Tech. Quality	Delivery Reliab.	Integration	Problem-Solving	Collaboration	Adaptability	Weighted /5
Achille Larregle	4.4	4.7	4.7	4.4	4.5	4.4	4.52
Eneko Orhategaray	4.7	4.1	4.2	4.3	4.1	4.4	4.33
Aiert Ceccon	4.4	4.0	4.0	4.2	4.3	4.2	4.20
Paul Brocvielle (self)	4.3	3.8	4.0	4.3	4.1	4.4	4.14

7.3 Individual analyses

Achille Larregle (4.52/5). Strongest coordination and final integration contribution. He kept deadlines visible, consolidated reports, and carried major Task 6/7 packaging work. Improvement area: delegation — some final integration work could have been shared earlier to reduce load concentration.

Eneko Orhategaray (4.33/5). Strong technical and visual contribution, especially in Task 6. His geometry, plotting, and quality-checking work made cooling-tower optimisation more understandable and defensible. Improvement area: making reasoning and ownership more visible while work is happening, not only in final outputs.

Aiert Ceccon (4.20/5). Contributed strongly in early optimisation tasks and helped maintain technical progress. Strengths were practical coding support, collaboration, and discussion leadership. Improvement area: clearer end-to-end ownership from intermediate technical work to final submission artifacts.

Self-assessment — Paul Brocvielle (4.14/5). Strengths: technical reasoning, problem framing, workflow setup, and adaptability. My setup and reproducibility work helped the project become easier to run and maintain. Improvement area: sustained visible ownership during high-pressure finalisation phases, especially in Tasks 6-7.

8. Conclusion

Task 1 predicted my direction reasonably well. I contributed most naturally through strategy, system thinking, questioning assumptions, and technical structure. The strongest evidence is in Tasks 3-4 and the reproducibility work around setup, README, dependency management, and repository cleanup.

The project also showed a clear development need: I must turn strategic input into owned, traceable deliverables more consistently, especially near deadlines. My value is highest when I combine conceptual engineering judgement with practical reproducibility and visible follow-through. For my future engineering career, the next step is to keep the creativity and adaptability of the ENTP profile, but pair it with stronger final-stage ownership and clearer handover discipline.

9. Timesheet Reconstruction Method

No formal weekly timesheet was maintained consistently, so the attached timesheet is a reconstructed estimate. I used three evidence sources: Task release/submission periods, repository commits, and memory of meeting and review activity. Each row has a confidence tag: high for weeks with direct artifact or commit evidence, medium for active windows with partial traceability, and low for broad planning or light review periods.

The objective is not artificial precision but a transparent account of when my effort was highest. The reconstruction shows higher effort during Tasks 3-4 and the reproducibility/setup period, with lighter support during Tasks 6-7.

Appendix Note

- Reconstructed timesheet: `appendices/paul_reconstructed_timesheet.csv`
- Claim-to-evidence matrix: `evidence/claim_to_evidence_matrix.csv`
- Contribution timeline: `evidence/contribution_timeline.md`
- Requirement traceability checklist: `evidence/task8_requirement_traceability_checklist.md`